Two-level normal-theory ML: full-information deviance, gradient, and the per-block contraction that magmaan already owns

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Purpose

Pin the math behind the magmaan Stream-C two-level ML core: the Muthén–Asparouhov full-information cluster deviance F, its analytic gradient $\nabla_\theta F$, the balanced-data (MUML) reduction used as the implementation unit test, and the unstructured saturated (H_1) score equations. The load-bearing result (Corollary 1) is that $\nabla_\theta F$ is the *ordinary per-block discrepancy-gradient contraction* magmaan already performs in `ml_value_gradient`: assemble two symmetric "weighted-residual" matrices G_W, G_B and a between-mean gradient, then contract them with the existing $d\,\mathrm{vech}(\Sigma)/d\theta$ and $d\mu/d\theta$ Jacobians. The two-level coupling lives entirely inside how G_W, G_B are built from the per-cluster-size sufficient statistics; no new model machinery is needed.

Maps to the Phase-0 contracts: F is `twolevel_value`, $\nabla_\theta F$ is `twolevel_value_gradient`, the H_1 solver is `twolevel_h1_moments`, all over `data::ClusterSampleStats`.

1 Model and notation

One group (groups are independent and add additively; restore a group index by summing F and stacking gradients with the per-group n_b/N weights, exactly as multi-group already does). Clusters $j = 1, \ldots, J$; cluster j holds n_j level-1 units; $N = \sum_j n_j$. Observed p-vector y_{ij} for unit i in cluster j. The two-level model decomposes $y_{ij} = \mu + y_{B,j} + y_{W,ij}$ with independent $y_{B,j} \sim N(0, \Sigma_B)$ (cluster-level) and $y_{W,ij} \sim N(0, \Sigma_W)$ (unit-level), so within cluster j

$$\mathrm{Cov}(y_{ij}, y_{i'j}) = \Sigma_B + \mathbf{1}\{i = i'\}\,\Sigma_W, \qquad \mathbb{E}[y_{ij}] = \mu.$$

Here $\Sigma_W = \Sigma_W(\theta)$, $\Sigma_B = \Sigma_B(\theta)$, $\mu = \mu(\theta)$ are the implied within covariance, between covariance, and (between-level) mean; in magmaan they are $\Sigma_W =$ `sigma[`b_W`]`, $\Sigma_B =$ `sigma[`b_B`]`, $\mu =$ `mu[`b_B`]` for the within/between blocks paired by `level_block_pairs`. v1 scope: a *shared* observed set, so Σ_W, Σ_B are both $p \times p$ over the same p variables.

Stack cluster j as $Y_j = (y_{1j}^\top, \ldots, y_{n_j j}^\top)^\top$. With $\mathbf{1} = \mathbf{1}_{n_j}$ and $J_{n_j} = \mathbf{1}\mathbf{1}^\top$,

$$\mathrm{Var}(Y_j) = V_j = I_{n_j} \otimes \Sigma_W + J_{n_j} \otimes \Sigma_B, \qquad \mathbb{E}[Y_j] = \mathbf{1} \otimes \mu. \tag{1}$$

Write the cluster mean $\bar{y}_j = \frac{1}{n_j}\sum_i y_{ij}$ and the within-cluster scatter $S_j^W = \sum_i (y_{ij} - \bar{y}_j)(y_{ij} - \bar{y}_j)^\top$.

2 The cluster deviance

Lemma 1 (Determinant/quadratic split)**.** *For V_j in* (1)*, with $M_{n_j} := (\Sigma_W + n_j \Sigma_B)^{-1}$,*

$$\log|V_j| = (n_j - 1)\log|\Sigma_W| + \log|\Sigma_W + n_j \Sigma_B|,$$
$$(Y_j - \mathbf{1}\otimes\mu)^\top V_j^{-1}(Y_j - \mathbf{1}\otimes\mu) = \mathrm{tr}(\Sigma_W^{-1} S_j^W) + n_j\,(\bar{y}_j - \mu)^\top M_{n_j}(\bar{y}_j - \mu).$$

Proof. The orthogonal split of $\mathbb{R}^{n_j}$ into the mean direction $u = n_j^{-1/2}\mathbf{1}$ and its complement diagonalizes V_j: J_{n_j} has eigenvalue n_j on u and 0 on $u^\perp$, while I_{n_j} is the identity on both. Hence V_j acts as $\Sigma_W + n_j\Sigma_B$ on $u \otimes \mathbb{R}^p$ (one copy) and as Σ_W on $u^\perp \otimes \mathbb{R}^p$ ($n_j - 1$ copies), giving the determinant. For the quadratic form, $(u^\top \otimes I)Y_j = \sqrt{n_j}\,\bar{y}_j$ carries the mean-direction part with covariance $\Sigma_W + n_j\Sigma_B$, contributing $n_j(\bar{y}_j - \mu)^\top M_{n_j}(\bar{y}_j - \mu)$; the complement carries the centered deviations with covariance Σ_W, contributing $\mathrm{tr}(\Sigma_W^{-1} S_j^W)$ since $\sum_i \|\cdot\|^2_{\Sigma_W^{-1}} = \mathrm{tr}(\Sigma_W^{-1} S_j^W)$. $\square$

Sum $-2\log L = \sum_j[\log|V_j| + (\cdot)^\top V_j^{-1}(\cdot)]$ over clusters and *group by distinct cluster size*. Let $\mathcal{D}$ be the set of distinct sizes, $J_d = \#\{j : n_j = d\}$, and define the sufficient statistics

$$\mathrm{SSW} = \sum_j S_j^W, \qquad T_d^{(1)} = \sum_{j:n_j=d} \bar{y}_j, \qquad T_d^{(2)} = \sum_{j:n_j=d} \bar{y}_j \bar{y}_j^\top. \tag{2}$$

These are exactly `ClusterGroupStats::within_scatter`, `ClusterSizePattern::sum_cluster_mean`, and `::sum_cluster_mean_cp` (Stream B). Note $\sum_j (n_j - 1) = N - J$.

Theorem 1 (Full-information two-level deviance)**.** *Dropping the additive constant* $Np\log(2\pi)$*, and with* $M_d := (\Sigma_W + d\Sigma_B)^{-1}$ *and the per-size centered cross-product*

$$A_d(\mu) := \sum_{j:n_j=d} (\bar{y}_j - \mu)(\bar{y}_j - \mu)^\top = T_d^{(2)} - \mu T_d^{(1)\top} - T_d^{(1)}\mu^\top + J_d\,\mu\mu^\top,$$

the model deviance is

$$F(\theta) = (N-J)\log|\Sigma_W| + \mathrm{tr}(\Sigma_W^{-1}\mathrm{SSW}) + \sum_{d\in\mathcal{D}} \Big[J_d \log|\Sigma_W + d\Sigma_B| + d\ \mathrm{tr}\big(M_d A_d(\mu)\big)\Big]. \tag{3}$$

Proof. Insert Lemma 1 into $-2\log L$; the within terms collect to $(N-J)\log|\Sigma_W| + \mathrm{tr}(\Sigma_W^{-1}\mathrm{SSW})$ using (2); the mean-direction terms with $n_j = d$ collect to $J_d \log|\Sigma_W + d\Sigma_B| + d\sum_{j:n_j=d}(\bar{y}_j - \mu)^\top M_d(\bar{y}_j - \mu)$, and the sum equals $\mathrm{tr}(M_d A_d(\mu))$. $\square$

Remark (Objective scale). magmaan minimizes $f = \frac{1}{2}F$ and reports the LRT statistic $T = F(\hat{\theta}) - F_{H_1}$ with F_{H_1} the converged saturated deviance (Section 5); $\mathrm{df} = p(p{+}1) + p - q$ with q the number of free θ (saturated count $= 2\cdot p(p{+}1)/2$ covariance parameters plus p between means). This mirrors the FIML convention: the optimizer carries the raw deviance, the saturated subtraction makes it a discrepancy.

3 Gradient

Let $W_0 := \Sigma_W^{-1}$ and define the per-size *between residual operator*

$$R_d := J_d\, M_d - d\, M_d A_d(\mu) M_d \qquad (\text{symmetric, } p\times p). \tag{4}$$

Theorem 2 (Matrix gradients)**.** *With* F *as in* (3)*,*

$$G_W := \frac{\partial F}{\partial \Sigma_W} = (N-J)\, W_0 - W_0\,\mathrm{SSW}\,W_0 + \sum_{d\in\mathcal{D}} R_d, \tag{5}$$

$$G_B := \frac{\partial F}{\partial \Sigma_B} = \sum_{d\in\mathcal{D}} d\, R_d, \tag{6}$$

$$g_\mu := \frac{\partial F}{\partial \mu} = -2\sum_{d\in\mathcal{D}} d\, M_d\big(T_d^{(1)} - J_d\,\mu\big). \tag{7}$$

Proof. Use $d \log |X| = \operatorname{tr}(X^{-1} dX)$ and $d \operatorname{tr}(X^{-1}A) = -\operatorname{tr}(X^{-1}(dX)X^{-1}A)$. For Σ_W: the within terms give $(N-J)W_0$ and $-W_0 \mathrm{SSW} W_0$; since $\partial(\Sigma_W + d\Sigma_B)/\partial\Sigma_W = I$, the d-sum gives $\sum_d [J_d M_d - d\, M_d A_d M_d] = \sum_d R_d$. For Σ_B: only the d-sum depends on Σ_B and $\partial(\Sigma_W + d\Sigma_B)/\partial\Sigma_B = d\, I$, scaling each bracket by d, giving $\sum_d d\, R_d$. For μ: only $A_d(\mu)$ depends on μ, and $\partial \operatorname{tr}(M_d A_d)/\partial\mu = -2M_d(T_d^{(1)} - J_d\mu)$, scaled by d and summed. □

Equations (5)–(6) expose the structure: R_d is the shared per-size between residual, weighted by 1 in G_W and by d in G_B. At a perfect between fit $A_d(\mu) = J_d(\Sigma_W + d\Sigma_B)$, so $R_d = 0$ for every d and the between channel is dormant in both gradients.

Corollary 1 (The contraction magmaan already owns)**.** *Let $\partial_\theta \operatorname{vech}\Sigma_W$, $\partial_\theta \operatorname{vech}\Sigma_B$ be the within/between blocks of the evaluator Jacobian `J_sigma`, and $\partial_\theta\mu$ the between block of `J_mu`. With the symmetric-vech reduction $v(G) := \operatorname{vech}(2G - \operatorname{diag} G)$ (off-diagonals doubled, the same map `ml_value_gradient` applies),*

$$\nabla_\theta F = (\partial_\theta \operatorname{vech}\Sigma_W)^\top v(G_W) + (\partial_\theta \operatorname{vech}\Sigma_B)^\top v(G_B) + (\partial_\theta\mu)^\top g_\mu.$$

Proof. $\partial F/\partial\theta_k = \operatorname{tr}(G_W\, \partial_{\theta_k}\Sigma_W) + \operatorname{tr}(G_B\, \partial_{\theta_k}\Sigma_B) + g_\mu^\top \partial_{\theta_k}\mu$ by the chain rule on Theorem 2; for symmetric $G, \partial\Sigma$ the trace equals $v(G)^\top \partial_{\theta_k} \operatorname{vech}\Sigma$. □

Implementation: `twolevel_value_gradient` forms $M_d, A_d, \mathrm{SSW}, W_0$ once, builds G_W, G_B, g_μ by (5)–(7), then reuses the existing per-block contraction. The only inputs beyond the single-level path are the cluster-size buckets; the Jacobian rows for the within block contract with $v(G_W)$, the between block with $v(G_B)$, and the between mean with g_μ.

4 Balanced reduction (the unit test)

Theorem 3 (MUML)**.** *If all $n_j = d$ (single distinct size), then with $S_{PW} := \mathrm{SSW}/(N-J)$, $\hat\mu_{H_1} = T_d^{(1)}/J$, and $C_B := A_d(\mu)/J$,*

$$F = \underbrace{(N-J)\big[\log|\Sigma_W| + \operatorname{tr}(\Sigma_W^{-1} S_{PW})\big]}_{\textit{within block}} + \underbrace{J\big[\log|\Sigma_W + d\Sigma_B| + \operatorname{tr}\big((\Sigma_W + d\Sigma_B)^{-1} C_B\big)\big]}_{\textit{between block}}.$$

Each bracket is a normal-theory ML fit term (constants dropped). Hence on balanced data F equals the two-block ML deviance of the `SampleStats` $\{(S_{PW}, N-J), (C_B, J)\}$ whose block-2 implied covariance is $\Sigma_W + d\Sigma_B$.

This is the gradient/value check for Stream C: build a balanced `ClusterSampleStats`, and assert `twolevel_value` and `twolevel_value_gradient` match `ml_value`/`ml_value_gradient` run on the two-block `SampleStats` above (with $\Sigma_2 = \Sigma_W + d\Sigma_B$), to machine precision; cross-check `twolevel_value_gradient` against a central finite difference of `twolevel_value` on unbalanced data.

5 Saturated (H_1) model

The saturated fit maximizes L over unstructured $(\Sigma_W, \Sigma_B, \mu)$; its score equations are $G_W = 0$, $G_B = 0$, $g_\mu = 0$ from Theorem 2. From $g_\mu = 0$,

$$\hat{\mu} = \Big(\sum_d d\, J_d\, M_d\Big)^{-1}\Big(\sum_d d\, M_d\, T_d^{(1)}\Big), \tag{8}$$

which for a single size reduces to $\hat{\mu} = T_d^{(1)}/J_d$ (the grand mean of cluster means). The covariance equations $G_W = G_B = 0$ are coupled and have no closed form for unbalanced $\mathcal{D}$; iterate (Fisher scoring on the three unstructured blocks, or two-level EM treating the cluster effects as missing) to convergence, the analogue of `fiml_h1_moments`. Balanced closed form (single d): $\hat{\Sigma_W} = S_{PW}$ and $\hat{\Sigma_B} = (C_B - S_{PW})/d$ when $C_B - S_{PW} \succeq 0$ (else the boundary $\hat{\Sigma_B} = 0$), with $\hat{\mu}$ as above; this seeds the iteration and gives a H_1 smoke test.

Remark (Start values). `twolevel_start_values`: take Σ_W from S_{PW} and Σ_B from the pooled $\bar{d}^{-1}(C_B - S_{PW})_+$ (averaging the per-size C_B), then read off the structural starts the single-level producers already compute from a covariance target (`fabin`/`simple`) applied per level.

References

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